

Technical Document #497: Database Systems - Comprehensive Overview

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ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Database indexing improves query performance by creating data structures that allow faster data retrieval.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Database caching stores frequently accessed data in memory. Redis and Memcached are popular caching solutions.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Database monitoring tracks performance metrics and identifies bottlenecks.
- Database connection pooling reuses database connections to reduce overhead.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.